

Purpose

A complete, implementation-ready specification an AI coding agent can follow to build Joe Kuntani's standalone digital brand platform. It covers the public website, booking funnel, portfolio, media kit, content CMS, campaign management, analytics, security, testing, deployment, and acceptance criteria.

Prepared for Neurodyne Corp

Document Map

Section	What the agent gets
1-3	Product intent, goals, users, constraints and success measures
4-7	Functional scope, journeys, modules and detailed requirements
8-11	Architecture, stack, data model and APIs
12-15	CMS, integrations, security and non-functional requirements
16-19	UX, analytics, testing and deployment
20-23	Implementation phases, tickets, acceptance criteria and definition of done

1. Product Summary

Build a premium, mobile-first digital presence for Joe Kuntani that converts attention into trust, enquiries, confirmed bookings, brand partnerships, media opportunities, and measurable commercial growth. The platform is not a social network and is not Creator OS. It is Joe's dedicated website and internal business management system.

Primary outcome

A brand or event organizer should be able to discover Joe, understand his value, review proof of work, select a service, submit a complete brief, and receive a professional response without relying on social-media direct messages.

2. Goals and Non-Goals

2.1 Goals

- Present Joe as a credible Ghanaian comedian, host, performer and brand partner.
- Centralize videos, press mentions, testimonials, audience metrics and brand work.
- Create a structured booking and sponsorship enquiry funnel.
- Allow Joe's team to manage public content without editing code.
- Track leads, proposals, bookings, campaigns, deliverables and payments.
- Provide measurable conversion and campaign-performance analytics.
- Lay a clean technical foundation that may later integrate with Creator OS without depending on it.

2.2 Explicit Non-Goals

- No multi-creator tenancy, creator marketplace or creator onboarding.
- No public fan social network, chat system or user-generated profile system.
- No complete accounting, payroll, merchandise fulfilment, public ticket-resale marketplace or multi-organizer ticketing platform in version 1. Joe's own event ticket sales are included.
- No automated contract execution or payment escrow in the MVP.
- No Creator OS branding in the public user experience.

3. Users, Roles and Permissions

Role	Description	Permissions
Public visitor	Fan, journalist, event organizer or brand representative	View all published content; download approved media kit; submit enquiries.
Administrator	Joe or authorized business manager	Full access to content, enquiries, campaigns, settings, users and analytics.
Content editor	Editor/social media manager	Create and edit content, portfolio, press, testimonials and media assets; cannot manage users or financial records.
Booking manager	Manager/agent	Manage enquiries, proposals, bookings, contacts, notes, tasks and status changes.
Analyst - optional	Read-only stakeholder	View dashboards and export reports; no write access.

4. Product Surfaces

Surface	Purpose
Public marketing website	Brand storytelling, social proof, portfolio discovery, media kit and conversion.
Booking and partnership forms	Collect structured, qualified enquiries.
Admin dashboard	Manage content, leads, bookings, campaigns, media assets and analytics.
Transactional email layer	Acknowledgements, internal notifications, follow-ups and status communications.
Public share pages	Share individual portfolio projects, campaigns, press features and event pages.
Event discovery and ticket checkout	Promote Joe's shows, display event banners and sell first-party tickets for events managed by Joe's team.

5. Public Website Information Architecture

Route	Page	Required content / behaviour
/	Home	Hero, short positioning, featured videos, service cards, audience proof, selected partners, testimonials, latest press, strong booking CTA.
/about	About Joe	Long biography, origin story, values, career timeline, achievements, professional portraits and downloadable short biography.
/work	Work / Portfolio	Filterable cards for comedy, campaigns, hosting, performances and

		collaborations.
/work/[slug]	Case study	Problem/context, Joe's role, creative concept, deliverables, media, results, testimonial and related work.
/videos	Videos	Featured and categorized videos using embedded YouTube/TikTok/Instagram links and poster images.
/services	Services	Brand campaigns, event hosting/MC, comedy performance, appearances, content production and custom partnerships.
/book	Book Joe	Multi-step enquiry form with service-specific questions and consent.
/media-kit	Media kit	Web summary plus controlled PDF download; metrics must show "last updated" date.
/press	Press & Media	Articles, interviews, podcasts, TV/radio appearances and downloadable press assets.
/contact	Contact	General contact form, approved business email and social links.
/privacy	Privacy notice	Data use, retention, cookies, enquiry handling and contact details.
/terms	Terms	Website terms, asset-use restrictions and booking disclaimer.
/events	Events	Upcoming and past events; featured banner, city/date filters, availability state and ticket CTA.
/events/[slug]	Event details	Hero/banner, event description, date/time, venue, map link, age/accessibility notes, ticket types, sales deadline, organizer terms and checkout CTA.
/tickets/[order-reference]	Ticket confirmation	Order summary, payment state, downloadable/mobile ticket access and support instructions. Must require a secure access token or authenticated order lookup.

6. Key User Journeys

6.1 Brand partnership enquiry

1. Visitor lands on Home or a campaign case study.
2. Visitor reviews audience proof, past partnerships and services.
3. Visitor selects "Start a Partnership".
4. Form dynamically requests campaign objective, dates, budget range, platforms, deliverables, usage rights, exclusivity and contact details.
5. System validates, stores the lead, sends an acknowledgement, and alerts the booking manager.

6. Manager reviews the lead, assigns an owner, adds notes/tasks and changes stage.
7. Manager creates or uploads a proposal and records the outcome.

6.2 Event booking

8. Organizer selects MC, comedy performance or appearance.
9. Form collects event type, date/time, venue, city/country, expected audience, duration, production requirements, budget and contact details.
10. System flags date conflicts against confirmed or tentative bookings.
11. Manager qualifies, schedules a call, issues a quote/proposal and updates the booking pipeline.

6.3 Content management

12. Editor signs in using MFA.
13. Editor creates a draft portfolio item or press feature.
14. Editor uploads/selects optimized media, adds SEO metadata and previews the page.
15. Editor submits or publishes according to permission.
16. System records an audit entry and invalidates relevant caches.

7. Functional Requirements

ID	Capability	Acceptance requirement
FR-001	Responsive public site	All public pages must work at 320px and above, with no horizontal overflow.
FR-002	Global CMS	Admin can edit navigation, footer, CTAs, contact details, social links and basic SEO settings.
FR-003	Portfolio management	CRUD, drafts, publish scheduling, categories, tags, featured state, image/video galleries, results and testimonials.
FR-004	Video library	Store metadata and external embed URLs; never upload large video files directly in MVP.
FR-005	Services	CRUD services, service-specific form questions, CTA text and display order.
FR-006	Booking form	Multi-step, save-in-progress in browser, server validation, spam prevention and confirmation reference number.
FR-007	Lead pipeline	Stages: New, Reviewing, Qualified, Call Scheduled, Proposal Sent, Negotiation, Won, Lost, Archived.
FR-008	Booking calendar	Month/list views; tentative and confirmed bookings; conflict warnings; iCal export.
FR-009	Campaign tracking	Track client, campaign, dates, deliverables, platform, status, fee, expenses, results and assets.
FR-010	Contacts/CRM	Organizations, contacts, communication history, notes, tasks and tags.
FR-011	Media kit	Generate web view from stored metrics

		and offer an admin-uploaded PDF version.
FR-012	Press and testimonials	CRUD, source attribution, publish date, quote approval and featured state.
FR-013	Asset library	Upload images/PDFs; folders/tags; alt text; dimensions; file size; usage references.
FR-014	Analytics	Traffic, conversions, enquiry sources, pipeline value, bookings won and content performance.
FR-015	Notifications	Email acknowledgement to enquirer; internal email to configured recipients; overdue task reminders.
FR-016	Exports	CSV export for enquiries, contacts, bookings and campaigns based on role.
FR-017	Audit log	Record sign-in, create, update, delete, publish, status and export events.
FR-018	Search	Admin global search across enquiries, contacts, campaigns, bookings and content.
FR-019	SEO	Per-page title, description, canonical URL, social image and structured data.
FR-020	Consent	Store enquiry privacy consent timestamp, version and source IP hash.
FR-020	Event management	Admin can create, edit, preview, publish, unpublish and cancel events with banners, venue, schedule, ticket-sales window, capacity and policy text.
FR-021	Ticket inventory	Support General Admission, VIP and configurable ticket types with price, currency, quantity, per-order limit, sale start/end and sold-out state.
FR-022	Ticket checkout	Customer selects ticket quantities, enters contact information, accepts event/refund terms and completes payment through a provider abstraction.
FR-023	Order and payment processing	Create idempotent pending orders, verify provider webhooks, confirm only verified payments, expire abandoned inventory holds and prevent overselling.
FR-024	Digital ticket issuance	Generate one unique signed QR ticket per admission unit and send a Resend confirmation email with secure ticket links.
FR-025	Ticket validation	Authorized staff can scan or manually enter ticket codes, see validity without exposing unnecessary personal data and

		mark tickets checked in once.
FR-026	Ticket operations	Admin can resend tickets, void tickets, record approved refunds, export attendee lists and view sales, revenue, fees and check-in progress.
FR-027	Event promotion banner	Admin can feature one or more active events as homepage banners with schedule, audience targeting dates and CTA.

8. Recommended Technical Architecture

Implementation assumption

Use a modular monolith for speed and maintainability. Keep clear domain boundaries so modules can later be integrated into Creator OS. Do not introduce microservices in version 1.

Layer	Recommendation
Frontend	Next.js 16 with App Router, TypeScript, React Server Components where appropriate, and a typed API client for the Go backend.
UI	Tailwind CSS + accessible headless components (Radix UI or shadcn/ui).
Backend API	Go 1.25+ modular monolith exposing versioned REST endpoints. Use chi or Gin, explicit domain services, request validation, structured errors and generated OpenAPI documentation.
Database	MongoDB Atlas using separate staging and production clusters/databases.
Data access	Official MongoDB Go Driver with repository interfaces, BSON models, schema validation rules and explicit index creation.
Authentication	Go backend-issued secure HTTP-only session cookies or short-lived access tokens with rotating refresh tokens; Argon2id password hashing and mandatory MFA for administrators.
Media storage	Cloudinary for images, PDFs and approved media assets using signed uploads, transformations, folders/tags and webhook verification.
Email	Resend for transactional email, using verified domains, reusable templates, idempotency keys and webhook-based delivery status tracking.
Analytics	PostHog for product analytics; Google Search Console for SEO; optional GA4.
Error monitoring	Sentry.
Hosting	Render Blueprint deployment: separate Next.js 16 web service and Go API service, with environment groups, health checks, preview/staging and production services. MongoDB runs on MongoDB Atlas.
Testing	Frontend: Vitest, React Testing Library and Playwright. Backend: Go testing package, testify and integration tests against an isolated MongoDB database.

CI/CD	GitHub Actions plus Render Blueprint: frontend lint/typecheck/tests/build, Go fmt/vet/staticcheck/tests/build, MongoDB index/schema validation and Playwright smoke tests.
-------	--

8.1 Required repository structure

```
joe-kuntani-platform/
├── app/
│   ├── (public)/
│   ├── admin/
│   └── api/
├── components/
│   ├── public/
│   ├── admin/
│   └── ui/
├── modules/
│   ├── auth/
│   ├── content/
│   ├── enquiries/
│   ├── crm/
│   ├── bookings/
│   ├── campaigns/
│   ├── media/
│   ├── analytics/
│   ├── notifications/
│   └── settings/
├── lib/
├── prisma/
├── public/
├── tests/
│   ├── unit/
│   ├── integration/
│   └── e2e/
├── docs/
└── .github/workflows/
```

9. Domain Model

Entity	Core fields
User	id, name, email, password_hash/null, role, mfa_enabled, status, last_login_at, created_at, updated_at
SiteSetting	key, JSON value, group, updated_by, updated_at
Page	id, slug, title, status, seo fields, published_at, content JSON
Service	id, name, slug, summary, description, category, active, sort_order, form_schema JSON
PortfolioItem	id, title, slug, category, summary, body, client, event_date, role, results JSON, featured, status, published_at
MediaAsset	id, type, storage_key, public_url, filename, mime_type, bytes, width, height, alt_text, tags JSON, uploaded_by
Video	id, title, platform, external_url, embed_url, thumbnail_asset_id, category, featured, published_at
PressItem	id, title, outlet, external_url, type, publish_date, excerpt, image_asset_id, featured
Testimonial	id, quote, person_name, title, organization, image_asset_id, approved, featured
Organization	id, name, industry, website, country, notes

Contact	id, organization_id, first_name, last_name, email, phone, job_title, source, tags
Enquiry	id, reference, type, service_id, organization_id, contact_id, stage, source, budget_min, budget_max, event/campaign dates, details JSON, consent fields, owner_id
EnquiryNote	id, enquiry_id, author_id, body, created_at
Task	id, title, due_at, status, priority, assignee_id, enquiry_id/campaign_id/booking_id
Booking	id, enquiry_id, title, service_id, start_at, end_at, venue, city, country, status, fee, currency, requirements JSON
Campaign	id, enquiry_id, organization_id, title, objective, start_date, end_date, status, fee, currency, expenses, results JSON
Deliverable	id, campaign_id, title, platform, format, due_at, status, published_url, approval_status
AudienceMetric	id, platform, metric_date, followers, reach, impressions, engagement_rate, audience JSON
AuditLog	id, actor_id, action, entity_type, entity_id, metadata JSON, created_at
Event	id, public_id, title, slug, summary, description, banner_asset_id, venue, address, city, country, timezone, starts_at, ends_at, doors_open_at, status, capacity, sales_start_at, sales_end_at, terms, refund_policy, featured, published_at
TicketType	id, event_id, name, description, price Decimal128, currency, quantity_total, quantity_reserved, quantity_sold, min_per_order, max_per_order, sales_start_at, sales_end_at, status, sort_order
TicketOrder	id, public_id, reference, event_id, buyer_name, buyer_email, buyer_phone, currency, subtotal, fees, total, status, payment_provider, payment_reference, idempotency_key, hold_expires_at, paid_at, refunded_at, created_at
TicketOrderItem	id, order_id, ticket_type_id, quantity, unit_price, line_total
IssuedTicket	id, public_id, order_id, order_item_id, event_id, ticket_type_id, attendee_name/null, qr_token_hash, status, checked_in_at, checked_in_by, voided_at, created_at
PaymentWebhook	id, provider, external_event_id, event_type, signature_valid, payload_hash, processing_status, processed_at, created_at
TicketCheckIn	id, ticket_id, event_id, checked_in_by, device_label/null, result, created_at

9.1 Data rules and indexes

- MongoDB documents use ObjectId as the internal _id and expose stable UUID public_id values where identifiers leave the API. All timestamps use UTC and are displayed in the configured business timezone.
- Public slugs are unique and immutable after publication unless a redirect is created.

- Use soft deletion for enquiries, contacts, bookings and campaigns; hard deletion only for unreferenced drafts/assets.
- Create MongoDB compound and unique indexes for enquiry stage/owner/created_at, event dates, normalized contact email, organization name, content status/published_at, public slugs and idempotency keys.
- Money uses MongoDB Decimal128 plus ISO currency code; never binary floating point.
- Sensitive free-text form data must not be included in analytics events or logs.

10. API Contract

Method	Endpoint	Purpose
GET	/api/public/site	Global published site configuration.
GET	/api/public/work	Published portfolio with filters and cursor pagination.
GET	/api/public/work/:slug	Published case study.
POST	/api/public/enquiries	Create enquiry; validate schema, consent and anti-spam token.
GET	/api/admin/enquiries	Authorized filtered list.
GET/PATCH	/api/admin/enquiries/:id	Read/update lead, owner and stage.
POST	/api/admin/enquiries/:id/notes	Add internal note.
GET/POST	/api/admin/bookings	List/create bookings.
PATCH	/api/admin/bookings/:id	Update booking and conflict-check.
GET/POST	/api/admin/campaigns	List/create campaigns.
PATCH	/api/admin/campaigns/:id	Update campaign.
POST	/api/admin/media/sign-upload	Issue signed upload request.
POST	/api/admin/media/complete	Persist upload metadata.
GET	/api/admin/analytics/overview	Dashboard aggregates for selected range.
GET	/api/admin/exports/:resource	Role-authorized CSV export.
GET	/api/public/events	List published upcoming events and featured banners.
GET	/api/public/events/:slug	Read published event, ticket types and live availability.
POST	/api/public/ticket-orders	Create an idempotent pending order and short inventory hold.
POST	/api/public/ticket-orders/:reference/checkout	Create provider checkout session for the pending order.
GET	/api/public/ticket-orders/:reference	Read payment/ticket state using secure access token.
POST	/api/webhooks/payments/:provider	Verify signature, deduplicate webhook and update order atomically.
GET/POST	/api/admin/events	List/create events.
GET/PATCH	/api/admin/events/:id	Read/update event and publication state.
GET/POST	/api/admin/events/:id/ticket-types	List/create ticket types.

PATCH	/api/admin/ticket-types/:id	Update inventory, pricing and sale state subject to sales constraints.
GET	/api/admin/ticket-orders	Filter orders by event, status, date and buyer.
POST	/api/admin/ticket-orders/:id/resend	Resend secure ticket email.
POST	/api/admin/tickets/validate	Validate and check in a QR/manual ticket code atomically.
POST	/api/admin/tickets/:id/void	Void a ticket with reason and audit event.
POST	/api/admin/ticket-orders/:id/refund	Request/record approved full or partial refund through provider abstraction.
GET	/api/admin/events/:id/attendees.csv	Role-authorized attendee export.

API rule

All mutation endpoints must validate with Zod, authorize by role and ownership, emit audit events, return consistent problem-detail errors, and use idempotency keys where duplicate submission is possible.

11. Booking Form Specification

Step	Fields
1. Intent	Enquiry type; selected service; how the visitor heard about Joe.
2. Contact	Full name; organization; role; business email; phone; country.
3. Details - event	Event type; date/time; venue; city/country; audience size; performance duration; production needs.
3. Details - brand	Campaign objective; target audience; channels; requested deliverables; usage rights; exclusivity; launch dates.
4. Commercial	Budget range; currency; decision deadline; additional notes.
5. Review	Summary; privacy consent; optional marketing consent; submit.

- Generate reference format JK-YYYY-XXXXXX.
- Prevent duplicate submissions using idempotency token and rate limiting.
- Use honeypot plus Turnstile/reCAPTCHA only when risk score requires it.
- Never expose Joe's private phone number or personal address.
- Acknowledgement must state that submission is not confirmation of availability or booking.

12. Admin Dashboard

Module	Required screens
Overview	KPIs, latest enquiries, upcoming bookings, overdue tasks, pipeline and recent content activity.
Enquiries	Table, filters, saved views, detail drawer/page, notes, tasks, stage history and proposal attachments.

Calendar	Month/week/list; tentative/confirmed/cancelled; conflict warnings.
Campaigns	Campaign list, detail, deliverables checklist, assets, performance and financial summary.
Contacts	Organizations and people with enquiry/campaign history.
Content	Pages, portfolio, videos, services, press, testimonials and preview.
Media	Asset grid/list, upload, metadata, filters, usage references and delete protection.
Analytics	Website, enquiries, pipeline, bookings, campaigns and audience metrics.
Settings	Brand details, emails, social links, SEO defaults, integrations, consent text and team access.
Audit	Searchable admin activity log for authorized admins.
Events	Event list, create/edit form, banner, schedule, venue, publication, ticket-sale window, capacity, policies and preview.
Tickets & Orders	Ticket types, inventory, orders, payments, refunds, ticket resend/void actions, attendee export and revenue summary.
Check-in	Mobile-first QR scanner/manual lookup, validity result, duplicate warning, offline-failure guidance and live attendance count.

13. Content and Brand Requirements

- The visual language must feel premium, playful and distinctly Ghanaian without relying on stereotypes.
- Use a restrained design system: 1 primary dark colour, 1 accent colour, neutral scale, consistent radii and spacing.
- Photography and video are the principal visual assets; decorative UI must not overpower Joe's work.
- Every public page must end with one context-appropriate CTA.
- All images require meaningful alt text; decorative images use empty alt text.
- Metrics and client logos may only be published after approval and must include a last-updated date where appropriate.
- No fake testimonials, logos, audience numbers or campaign results. Seed data must be clearly marked as placeholder in non-production environments.

14. Integrations

Integration	MVP behaviour	Fallback
Email provider	Send enquiry acknowledgement and internal alerts.	Persist failed send and retry with backoff.
Object storage	Direct signed uploads; optimized public delivery.	Reject uploads when storage unavailable; preserve draft.
YouTube/Instagram/TikTok	Store external URLs and approved embeds.	Display poster + outbound link when embedding fails.

Analytics	Capture page view, CTA click, form step, submission and download events.	Server-side conversion log remains authoritative.
Calendar	iCal export; optional Google Calendar sync later.	Internal booking calendar is source of truth.
PDF media kit	Admin uploads approved PDF.	Web media-kit page remains accessible.
Payment provider	Provider adapter creates checkout sessions; signed webhook is authoritative for payment confirmation and refunds.	Order remains Pending/Payment Failed; inventory hold expires automatically.
QR ticket delivery	Generate signed opaque ticket tokens; Resend sends secure links and confirmation.	Admin can resend; customer can retrieve through secure order page.

15. Security, Privacy and Compliance

- Mandatory MFA for Administrator; recommended for all staff.
- Role-based authorization enforced server-side on every protected action.
- Passwords hashed with Argon2id or a trusted authentication provider.
- Secure cookies: HttpOnly, Secure, SameSite=Lax/Strict as appropriate.
- CSRF protection for state-changing browser requests.
- Rate limiting for authentication, enquiry submission, export and upload endpoints.
- File type allowlist, size limits and malware scanning where available.
- Signed URLs for private proposal and contract attachments.
- Audit all exports and high-risk administrative changes.
- Database backups with tested restoration procedure.
- Privacy notice must state purpose, retention and contact mechanism. Default enquiry retention: 24 months after last activity unless legally required otherwise.
- Provide admin actions to export or delete a contact's personal data, subject to legitimate record-retention needs.

16. Non-Functional Requirements

Area	Target
Performance	Public LCP <= 2.5 seconds at 75th percentile on mobile; optimized images; minimal third-party scripts.
Availability	99.5% monthly target excluding planned maintenance.
Accessibility	WCAG 2.2 AA target; keyboard access, visible focus, labels, contrast, reduced-motion support.
SEO	Server-rendered metadata, sitemap, robots, canonical URLs, Open Graph and structured data.
Scalability	Support 100k monthly visits and 10k stored enquiries without redesign.
Observability	Structured logs, request IDs, error tracking, uptime checks and alerting.
Maintainability	Strict TypeScript in Next.js, idiomatic Go with clear domain boundaries, no business logic in React components, and

	documented MongoDB schema/index changes.
Compatibility	Latest two major versions of Chrome, Safari, Edge and Firefox; iOS/Android mobile browsers.
Localization readiness	All public strings centralized; English first; architecture must allow future language additions.

17. Analytics and Events

Event	Properties
page_view	path, referrer, campaign parameters
cta_clicked	cta_id, page, destination
service_viewed	service_slug
work_viewed	work_slug, category
media_kit_downloaded	source_page, version
booking_started	service, source
booking_step_completed	step, service
booking_submitted	reference, enquiry_type, source
booking_failed	step, error_code
admin_stage_changed	from, to, enquiry_id - internal only
campaign_completed	campaign_id, result summary - internal only
event_viewed	event_slug, source
ticket_selection_started	event_id, ticket_type_ids
ticket_checkout_started	order_reference, event_id, quantity
ticket_purchase_completed	order_reference, event_id, amount, currency, quantity
ticket_purchase_failed	order_reference, event_id, error_code
ticket_checked_in	event_id, ticket_type_id - internal only

Privacy rule

Do not send names, email addresses, phone numbers, free-text notes or confidential campaign details to third-party analytics tools.

18. Testing Strategy

Test layer	Minimum coverage
Unit	Validation schemas, permissions, conflict detection, reference generation, money calculations and status transitions.
Integration	Database repositories, enquiry creation, email outbox, media completion, exports and audit logging.
Component	Forms, tables, filters, dialogs, accessible navigation and error states.
E2E	Public enquiry journey; admin sign-in; lead qualification; booking creation; content publish; role restrictions.

Security	Rate limit, unauthorized access, IDOR attempts, unsafe uploads, XSS payloads and session invalidation.
Performance	Lighthouse CI on key public routes; basic load test for enquiry endpoint.
Accessibility	Automated axe checks plus manual keyboard and screen-reader smoke test.
Ticketing unit	Inventory availability, hold expiry, price totals, per-order limits, QR token verification and ticket/order state transitions.
Ticketing integration	Concurrent order creation, atomic stock reservation, webhook idempotency, payment confirmation, ticket issuance, refund/void flow and check-in uniqueness.
Ticketing E2E	Event discovery to checkout; successful webhook to ticket email; sold-out state; failed payment release; ticket scan; duplicate scan warning; admin attendee export.

18.1 Required E2E scenarios

17. Brand visitor submits a valid partnership enquiry and receives acknowledgement.
18. Invalid or bot submission is rejected without creating a duplicate lead.
19. Booking manager sees the new lead, assigns it, adds a note and moves it to Qualified.
20. Manager creates a tentative booking and receives a warning for an overlapping confirmed booking.
21. Editor publishes a case study; it appears publicly with correct SEO metadata.
22. Content editor is denied access to user management and financial exports.
23. Administrator disables a user and existing sessions are revoked.
24. Failed email remains visible to admin and retries successfully.

19. Deployment and Environments

- Environments: local, preview per pull request, staging, production.
- Never share production databases, storage buckets or email recipients with staging.
- Use environment validation at startup; application must fail fast when required secrets are missing.
- Run backward-compatible MongoDB schema and index change scripts as a controlled release step before application rollout.
- Enable MongoDB Atlas continuous backups or daily snapshots with tested retention and restoration procedures.
- Configure Render custom domains and TLS, www redirect, Resend domain authentication (SPF/DKIM/DMARC), Cloudinary signed-upload credentials, error monitoring and uptime checks.
- Production content must be entered through CMS or approved import script; do not hardcode client claims.

19.1 Required Render Blueprint services

Service	Configuration
joe-web	Next.js 16 web service. Build the frontend, set API base URL, expose health route and deploy from the monorepo web directory.

joe-api	Go web service. Build a static Linux binary, expose /health/live and /health/ready, and connect to MongoDB Atlas, Resend and Cloundinary.
Environment groups	Separate staging and production secrets. Never reuse MongoDB databases, Resend recipients or Cloundinary folders across environments.
render.yaml	Define both services, build/start commands, health checks, branch strategy, auto-deploy policy and required environment variables.
Release process	Validate MongoDB indexes/schema scripts, deploy API, run health checks, then deploy the Next.js web service and execute Playwright smoke tests.

20. Implementation Phases

Phase	Time	Exit outcome
Phase 0 - Discovery and assets	3-5 days	Confirm brand positioning, services, contact workflow, approved biography, images, videos, logos, testimonials, metrics and legal text. Produce content inventory and missing-assets list.
Phase 1 - Foundation	Week 1	Initialize the monorepo, Next.js 16 frontend, Go API, design tokens, CI, MongoDB Atlas, auth, roles, observability and Render staging deployment.
Phase 2 - Public site	Weeks 2-3	Build Home, About, Services, Work, Videos, Press, Media Kit, Contact, SEO and responsive navigation.
Phase 3 - Enquiries and CRM	Weeks 4-5	Build multi-step forms, validation, email notifications, enquiries, contacts, organizations, notes, tasks and pipeline.
Phase 4 - Bookings and campaigns	Weeks 6-7	Calendar, conflict detection, campaign/deliverable tracking, proposal attachments and exports.
Phase 5 - CMS and assets	Weeks 8-9	Content modules, draft/publish workflow, media library, previews, permissions and audit log.
Phase 6 - Analytics and hardening	Week 10	Dashboards, event tracking, performance, accessibility, security review, backup/restore and release checklist.
Phase 7 - Launch	Week 11	Content migration, final QA, redirects, monitoring, training, production launch and post-launch verification.
Phase 4A - Events and ticketing	Weeks 6-8	Event CMS, banners, ticket types, inventory reservation, payment adapter/webhooks, QR ticket issuance, order management, attendee exports and mobile check-in.

21. AI Agent Execution Rules

Agent directive

Implement one vertical slice at a time. Every slice must include schema/migration, service logic, authorization, UI, tests, telemetry and documentation. Do not mark a feature complete when only the interface exists.

- Read this specification and create /docs/implementation-plan.md before coding.
- Create GitHub issues or a local task checklist mapped to the work packages below, separating Next.js frontend, Go API and infrastructure work.
- Use small, reversible MongoDB schema/index change scripts and seed only non-sensitive demo data.
- Never invent Joe's biography, statistics, clients, testimonials, pricing or contact information. Use explicit placeholders and an admin "content incomplete" warning.
- Keep public content separate from application code.
- Enforce authorization in server code, not only by hiding UI controls.
- Add tests before closing each ticket and run the full quality gate before every phase handoff.
- Update README, environment example, API documentation and architecture decisions as implementation changes.
- When a requirement is ambiguous, choose the simplest secure implementation consistent with the product goals and record the decision in /docs/decisions/.

21.1 Initial ticket backlog

Ticket	Outcome
JK-001	Bootstrap Next.js 16 frontend, Go backend, shared OpenAPI contracts, linting, tests and CI.
JK-002	Create MongoDB collections, validation rules, indexes and seed framework using the official Go driver.
JK-003	Implement staff authentication, MFA, roles and session revocation.
JK-004	Implement design system, public shell and admin shell.
JK-005	Implement global settings and SEO configuration.
JK-006	Implement services and service-specific form schemas.
JK-007	Implement portfolio, videos, press and testimonials.
JK-008	Implement media upload, metadata and asset selection.
JK-009	Implement public booking/partnership form and acknowledgement email.
JK-010	Implement organizations, contacts and enquiry pipeline.
JK-011	Implement notes, tasks, stage history and internal notifications.
JK-012	Implement booking calendar and conflict detection.
JK-013	Implement campaigns and deliverables.
JK-014	Implement dashboards and privacy-safe analytics events.
JK-015	Implement CSV exports and audit logging.
JK-016	Complete accessibility, performance, security and browser QA.
JK-017	Create deployment, backup, restore and incident runbooks.
JK-018	Migrate approved content and execute production launch

	checklist.
JK-019	Implement Event and TicketType MongoDB collections, validation rules, indexes and admin CRUD.
JK-020	Implement public events list/detail pages and scheduled homepage event banners.
JK-021	Implement ticket inventory calculations, atomic reservation holds and expiry worker.
JK-022	Implement payment-provider interface, checkout session creation, signed webhook verification and idempotency.
JK-023	Implement paid order confirmation, unique QR ticket generation and Resend ticket delivery.
JK-024	Implement ticket orders admin, resend, void, refund records and attendee CSV export.
JK-025	Implement mobile-first ticket scanner/manual validation and one-time atomic check-in.
JK-026	Add ticket sales, revenue, inventory and attendance dashboards plus security/load tests.

22. Release Acceptance Criteria

- All routes in Section 5 are complete, responsive and populated with approved content or clearly marked placeholders in staging only.
- A visitor can submit each enquiry type and the record appears once in admin with correct source and consent data.
- Internal recipients receive notifications and the visitor receives a professional acknowledgement.
- Booking manager can process a lead from New to Won/Lost with notes, tasks, proposal files and full stage history.
- Booking conflicts are detected and visible before confirmation.
- Editor can create, preview, publish, unpublish and update content without deployment.
- Role restrictions pass automated and manual tests.
- No critical/high security findings remain open.
- Key public pages meet accessibility and performance targets.
- Backups, restoration, monitoring and incident contact procedures are documented and tested.
- Joe or his authorized manager has reviewed and approved the production content and public claims.

23. Definition of Done

The product is done when

Joe has a fast, credible public platform that presents his brand professionally; visitors can submit structured commercial enquiries; his team can manage content and opportunities securely; performance can be measured; and the entire system is deployed, monitored, documented, tested and maintainable without Creator OS.

24. Event Ticketing Module

This module allows Joe's team to promote and sell tickets for Joe Kuntani events directly on the platform. It is first-party event commerce for Joe's own shows, not an open marketplace where unrelated organizers or users can list or resell tickets.

24.1 Core customer journey

- Visitor sees a scheduled event banner on the homepage or opens the Events page.
- Visitor opens the event detail page and reviews the venue, date, policies and available ticket types.
- Visitor selects quantities. The API validates limits and creates a short inventory hold.
- Visitor supplies contact details, accepts the event/refund terms and starts payment.
- The payment provider webhook - not the browser redirect - confirms payment.
- The backend atomically marks the order Paid, converts reserved inventory to sold inventory and issues unique QR tickets.
- Resend delivers a purchase confirmation and secure ticket-access link.
- At the venue, authorized staff scan each QR code. The API validates event, signature, state and prior check-in before recording admission.

24.2 Event lifecycle

Draft -> Scheduled -> On Sale -> Sales Closed -> Completed. Cancelled may be entered from any pre-completion state by an authorized manager. Publishing and sale state are separate: an event may be publicly visible before ticket sales open. Cancellation must stop checkout immediately, preserve records and display approved customer guidance.

24.3 Order, payment and ticket states

Order states: Pending, Awaiting Payment, Paid, Payment Failed, Expired, Partially Refunded, Refunded, Cancelled. Ticket states: Active, Checked In, Voided, Refunded. State changes must be implemented through domain methods with audit records; clients must not set arbitrary states.

24.4 Inventory and overselling controls

- Use an atomic MongoDB transaction or conditional update to reserve inventory only when enough units remain.
- Pending checkout holds expire after a configurable period, recommended 10 minutes.
- Webhook processing must be idempotent by provider event ID and payment reference.
- Never issue tickets from a success URL or unverified client callback.
- Reconciliation must flag paid provider transactions that do not have a completed local order.
- For a payment confirmed after hold expiry, attempt a safe reconciliation; never silently oversell.

24.5 Ticket and QR security

- QR codes contain an opaque random token or signed compact payload; they must not expose buyer email, phone or sequential database IDs.
- Store only a secure hash of bearer-style random tokens where possible.
- Check-in uses an atomic transition from Active to Checked In so two devices cannot admit the same ticket.

- Scanner responses show only the minimum information required: validity, ticket type, event and masked attendee/order context.
- All resend, void, refund, export and manual override actions require authorization and audit logging.

24.6 Payment provider abstraction

The Go backend must expose a `PaymentProvider` interface so the initial provider can be changed without rewriting ticketing logic. Required operations: `CreateCheckout`, `VerifyWebhook`, `GetPaymentStatus` and `Refund`. Provider secrets exist only in the Go API. The final provider and Ghana payment methods must be confirmed during implementation; do not hardcode a vendor-specific domain model.

24.7 Email requirements

Use Resend templates for payment confirmation, ticket delivery, payment failure where useful, event reminder, event cancellation and refund confirmation. Emails must include Joe's approved support address, event policies, order reference and secure links. Do not attach large files when secure web tickets are sufficient.

24.8 Admin reporting

For each event show gross sales, refunds, net recorded revenue, paid orders, tickets sold by type, inventory remaining, conversion rate where available, check-ins and attendance percentage. Financial values must distinguish platform-recorded totals from payment-provider settlement data.

24.9 Ticketing acceptance criteria

- An authorized editor can publish an event and schedule its homepage banner without deployment.
- A customer can buy one or more available ticket types on mobile and receive a secure digital ticket after verified payment.
- Two concurrent buyers cannot purchase more tickets than available inventory.
- Duplicate payment webhooks do not create duplicate orders, tickets or emails.
- Failed or abandoned payments release inventory after the hold expires.
- A valid active ticket checks in once; a second scan returns an obvious duplicate-already-used result.
- Cancelled, voided, refunded and wrong-event tickets cannot be admitted.
- Authorized staff can export attendees and see sales/check-in reports; unauthorized roles are denied.
- Every price, refund and check-in mutation is audited.
- The ticket flow passes accessibility, mobile, security and load tests before launch.